



DOC ID: SL-POP-ERP-SUMMARY-001

Consolidated Release: October 2026 • Ver 1.0 Final

MASTER EXECUTIVE PORTFOLIO & FINANCIAL ROADMAP

Popular Auto Spare ERP — Complete 9-Module Master Milestone & Budget Summary

Executive consolidated milestone breakdown, multi-module timeline mapping, engineering resource matrix, and grand commercial investment schedule for Popular Auto Spare & A/C Parts Co. W.L.L. Prepared by SaNDS Lab Middle East W.L.L in strategic alignment with UniGlobal Consultancy.

TOTAL MODULES

9 Modules

End-to-End Automotive ERP

TOTAL INVESTMENT

BD 34,090.910

Fixed Baseline in Bahraini Dinars

CUMULATIVE EFFORT

88 Weeks

17.6 Engineering Months

DEDICATED TEAM

5 Specialists

100% Dedicated Full-Time

VERIFICATION TRANCHES

35 Gates

Zero-Risk Milestones



9 / 9

Modules Documented

Complete functional & technical specs



BD 1,363.636

Monthly Team Burn-Rate

Fixed transparent rate card



100% Multi-Branch

Enterprise Coverage

Warehouses, POS counters, mobile floor



15-Day UAT

Acceptance Grace SLA

Rigorous milestone verification



Total Fixed Project Investment (All 9 Modules)

Unified contract value covering the complete design, engineering, multi-branch testing, data migration, hardware integrations, training, and production deployment across all 9 modules for Popular Auto Spare & A/C Parts Co. W.L.L.

BD 34,090.910

THIRTY-FOUR THOUSAND NINETY BAHRAINI DINARS & 910 FILS

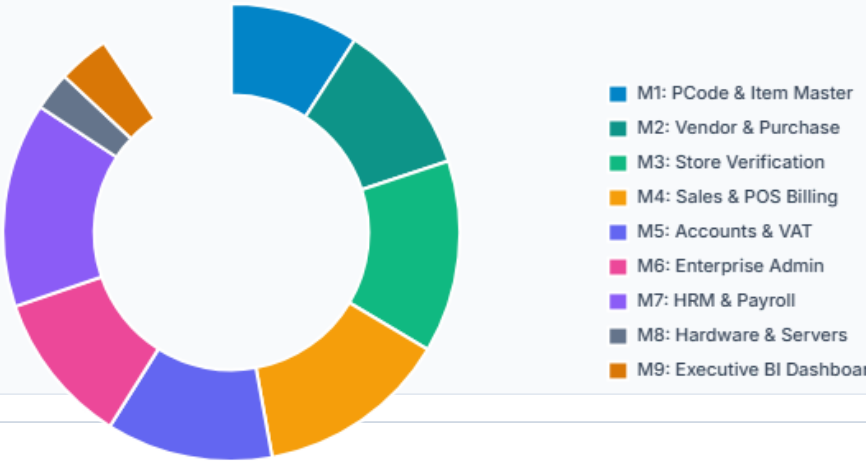
Visual Milestone & Budget Analytics

Visualizing budget distribution and engineering effort allocation across all 9 modules

INTERACTIVE VISUAL INSIGHTS

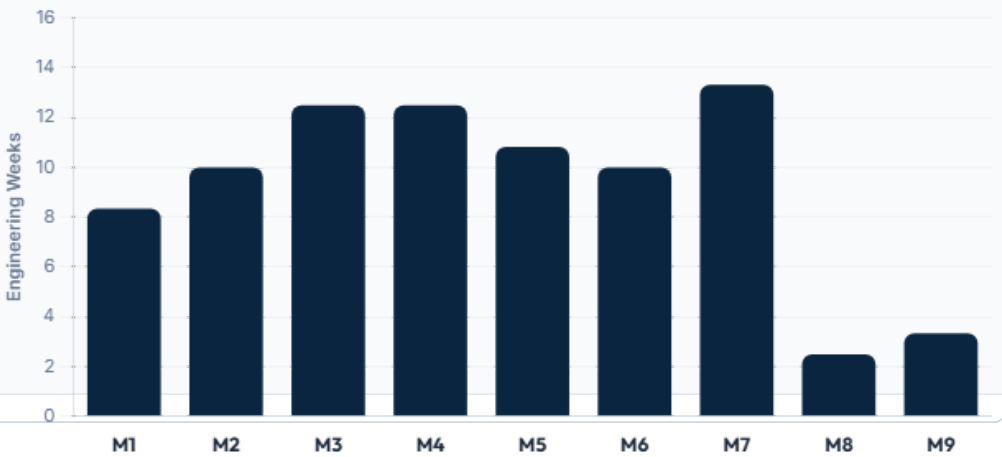
Budget Distribution by Module (BHD)

Total: BD 34,090.910



Engineering Timeline by Module (Weeks)

Total: 88 Working Weeks



 Master 9-Module Budgeting & Milestone Matrix (BHD)

At-a-glance comparative overview of all 9 modules, timelines, effort, deliverables, and total fees in Bahraini Dinars

ALL AMOUNTS IN BAHRAINI DINARS (BHD)

#	Module Scope & Implementation Domain	Timeline	Effort	Milestone Gates	Total Cost (BHD)	Weekly Burn	Share (%)	Action
1	SL-P0P-ERP-MS-001 BA Ref: DOC-001 v1.0 PCode Generation, Cataloguing & Item Master Engine	10 Weeks	2.50 Mo	4 Gates (25%)	BD 3,409.091	BD 340.909	10.00%	
2	SL-P0P-ERP-MS-002 BA Ref: DOC-002 v1.0 Vendor & Purchase Management, Supplier Portal & 3-Way Match	12 Weeks	3.00 Mo	4 Gates (25%)	BD 4,090.909	BD 340.909	12.00%	
3	SL-P0P-ERP-MS-003 BA Ref: DOC-003 v1.0 Store Verification, Multi-Warehouse Stock & Location Matrix	15 Weeks	3.75 Mo	4 Gates (25%)	BD 5,113.636	BD 340.909	15.00%	
4	SL-P0P-ERP-MS-004 BA Ref: DOC-004 v1.0 Sales Process, Mobile POS, Multi-Branch Billing & Return Control	15 Weeks	3.75 Mo	4 Gates (25%)	BD 5,113.636	BD 340.909	15.00%	
5	SL-P0P-ERP-MS-005 BA Ref: DOC-005 v1.0 Accounting & Financial Management, General Ledger, AP/AR & VAT	13 Weeks	3.25 Mo	4 Gates (25%)	BD 4,431.818	BD 340.909	13.00%	
6	SL-P0P-ERP-MS-006 BA Ref: DOC-006 v1.0 Enterprise Administration, Facility, Fixed Assets, Fleet & Vault	12 Weeks	3.00 Mo	4 Gates (25%)	BD 4,090.909	BD 340.909	12.00%	
7	SL-P0P-ERP-MS-007 BA Ref: DOC-007 v1.0 Human Resource Management, Biometrics, Leave, Payroll & Gratuity	16 Weeks	4.00 Mo	4 Gates (25%)	BD 5,454.548	BD 340.909	16.00%	
8	SL-P0P-ERP-MS-008 BA Ref: ARCH-001 v1.0 Hardware Setup, Handheld QR Devices, ESC/POS & Cloud Cluster	3 Weeks	0.75 Mo	3 Gates (33%)	BD 1,022.727	BD 340.909	3.00%	
9	SL-P0P-ERP-MS-009 BA Ref: DOC-009 v1.0 Executive Management Dashboard, 8-Module BI & Mobile Cockpit	4 Weeks	1.00 Mo	4 Gates (25%)	BD 1,363.636	BD 340.909	4.00%	
GRAND TOTAL: COMPLETE 9-MODULE AUTOMOTIVE ERP PORTFOLIO		88 Weeks	17.60 Mo	35 Gates	BD 34,090.910	BD 387.397	100.00%	Full Suite



9-Module Architectural Scope & Deliverable Pillars

Key functional scope, core modules, and business impact for Popular Auto Spare Co.

FUNCTIONAL PORTFOLIO

MODULE 1

BD 3,409.091

PCode Generation & Universal Item Master

Automated smart PCode generation algorithms, multi-brand hierarchy (Car Make/Model/Year), 1-to-Many OEM cross-reference matrix, 50,000+ SKU legacy data cleansing, and sub-100ms multi-parameter fuzzy search.

🕒 10 Weeks (2.5 Mo)

MODULE 2

BD 4,090.909

Vendor & Purchase Management with 3-Way Match

Multi-country vendor master, tokenized supplier RFQ portal, MOQ reorder buffering, CTO purchase approvals, automated PO splits by brand, Goods Receipt Note (GRN) with landed cost, and 3-way invoice matching.

🕒 12 Weeks (3.0 Mo)

MODULE 3

BD 5,113.636

Store Verification & Multi-Warehouse Stock Control

5-tier warehouse spatial layout (Zone/Aisle/Rack/Shelf/Bin), handheld Android scanner engine, blind 60/40 cycle counts, inter-branch stock transfers (GIT) with driver signatures, and FIFO inventory valuation.

🕒 15 Weeks (3.75 Mo)

MODULE 4

BD 5,113.636

Sales Process, Mobile POS & Branch Cash Control

High-speed counter POS, mobile Android handheld floor sales, dynamic QR cart handoff, credit customer credit-risk limits, verified invoice returns with condition grading, and End-of-Day cash drawer closing locks.

🕒 15 Weeks (3.75 Mo)

MODULE 5

BD 4,431.818

Accounting, General Ledger, AP/AR & VAT Compliance

5-group Chart of Accounts (COA), real-time double-entry GL, automated AP/AR subledgers, multi-currency treasury, automated Bank Reconciliation (BRS), PDC management, and Bahrain NBR 10% VAT return filings.

🕒 13 Weeks (3.25 Mo)

MODULE 6

BD 4,090.909

Enterprise Administration, Fixed Assets & Fleet

Centralized branch facility ticketing, fixed asset QR tagging with automated straight-line depreciation, company vehicle fleet logs with Total Cost of Ownership (TCO), and legal document expiry alerts (CR, Leases).

🕒 12 Weeks (3.0 Mo)

MODULE 7

BD 5,454.548

HR Management, Biometrics, Payroll & Gratuity

Employee 360 master, physical biometric clock-in & geofenced mobile attendance, Bahrain Labour Law leave rules, automated monthly payroll engine, SIO/LMRA/WPS bank export, and End-of-Service Benefit (EOSB) settlements.

🕒 16 Weeks (4.0 Mo)

MODULE 8

BD 1,022.727

Hardware Setup, Scanners, Printers & Cloud Cluster

3-tier cloud server architecture (Dev / Staging / Prod), RabbitMQ asynchronous background queue, Android PDA scanner listeners, ESC/POS thermal receipt & barcode label printers, and biometric sync daemons.

🕒 3 Weeks (0.75 Mo)

MODULE 9

BD 1,363.636

Executive Management Dashboard & BI Cockpit

OLAP data warehouse aggregation engine, real-time KPI metrics across all 8 modules, automated anomaly detection, daily 7:00 AM WhatsApp/Email executive digest, and secure C-Suite mobile executive reporting.

🕒 4 Weeks (1.0 Mo)

Complete 35 Milestone Tranches Detailed Breakdown (BHD)

Specific milestone gates, percentages, exact BHD allocations, and payment triggers for each phase

35 MILESTONE TRANCHES

MOD 1 Module 1: PCode Generation & Item Master Engine (10 Weeks)

BD 3,409.091

MILESTONE 1.1 (25%)

BD 852.273

PCode DB Schema, Data Cleansing Engine & Base Generation Algorithm.

📅 Weeks 1-3

MILESTONE 1.2 (25%)

BD 852.273

Brand / Model / Spec Dynamic Hierarchy & Automated PCode Generator Engine.

📅 Weeks 4-5

MILESTONE 1.3 (25%)

BD 852.273

Universal Cross-Reference / Interchange Matrix & Barcode Tagging Engine.

📅 Weeks 6-8

MILESTONE 1.4 (25%)

BD 852.273

Real-Time Search, Legacy Migration (50,000+ SKUs) & Multi-Branch Final UAT.

📅 Weeks 9-10

MOD 2 Module 2: Vendor & Purchase Management (12 Weeks)

BD 4,090.909

MILESTONE 2.1 (25%)

BD 1,022.727

Supplier Master, Tokenized Vendor Portal & Automated RFQ Engine.

📅 Weeks 1-3

MILESTONE 2.2 (25%)

BD 1,022.727

Multi-Vendor Quotation Evaluation Matrix & Automated PO Split Engine.

📅 Weeks 4-6

MILESTONE 2.3 (25%)

BD 1,022.727

Goods Receipt Note (GRN), Multi-Currency Landed Cost & QC Inspection.

📅 Weeks 7-9

MILESTONE 2.4 (25%)

BD 1,022.727

3-Way Invoice Matching, Vendor SLA Scorecards & Multi-Branch Sign-Off.

📅 Weeks 10-12

MOD 3 Module 3: Store Verification & Stock Control (15 Weeks)

BD 5,113.636

MILESTONE 3.1 (25%)

BD 1,278.409

Warehouse Spatial Hierarchy (5 Tiers), Bin Matrix & Inward PVN Setup.

📅 Weeks 1-4

MILESTONE 3.2 (25%)

BD 1,278.409

Handheld Scanner Engine, Blind 60/40 Cycle Counts & Variance Alerts.

📅 Weeks 5-8

MILESTONE 3.3 (25%)

BD 1,278.409

Inter-Branch Stock Transfers (GIT), Multi-Tier Discrepancy & FIFO Valuation.

📅 Weeks 9-12

MILESTONE 3.4 (25%)

BD 1,278.409

Min/Max Safety Stock Auto-Replenishment, Aging & Final Warehouse Sign-Off.

📅 Weeks 13-15

MOD 4 Module 4: Sales Process, POS & Branch Financial Control (15 Weeks)

BD 5,113.636

MILESTONE 4.1 (25%)

BD 1,278.409

Multi-Branch POS Architecture, Offline Sync & Cash Drawer Session Control.

📅 Weeks 1-4

MILESTONE 4.2 (25%)

BD 1,278.409

Counter & Mobile Handheld POS Billing, Multi-Price Matrix & Barcodes.

📅 Weeks 5-8

MILESTONE 4.3 (25%)

BD 1,278.409

Credit Customer Governance, Credit Limit Locks, Delivery Notes & B2B Invoices.

📅 Weeks 9-12

MILESTONE 4.4 (25%)

BD 1,278.409

Verified Sales Returns, End-of-Day Branch Reconciliation & Audit UAT.

📅 Weeks 13-15

MOD 5 Module 5: Accounting & Financial Management (13 Weeks)

BD 4,431.818

MILESTONE 5.1 (25%)

BD 1,107.955

Dynamic Chart of Accounts (COA), Cost Centers & Auto Journal Engine.

MILESTONE 5.2 (25%)

BD 1,107.955

AP / AR Subledgers, Aging Analysis, Statements & Credit Lock Rules.

MILESTONE 5.3 (25%)

BD 1,107.955

Multi-Currency Treasury, Automated Bank Reconciliation & Cashflow Engine.

MILESTONE 5.4 (25%)

BD 1,107.955

Bahrain NBR 10% VAT Return Generator, Balance Sheet, P&L & Final Sign-Off.

📅 Weeks 1–3

📅 Weeks 4–6

📅 Weeks 7–10

📅 Weeks 11–13

MOD 6 **Module 6: Enterprise Administration & Fixed Assets (12 Weeks)**

BD 4,090.909 ▾

MILESTONE 6.1 (25%)

BD 1,022.727

Facility Management, Maintenance Ticketing, Utility Logs & Lease Alerts.

📅 Weeks 1–3

MILESTONE 6.2 (25%)

BD 1,022.727

Fixed Asset Master, QR Tagging, Custodian Tracking & Depreciation Engine.

📅 Weeks 4–6

MILESTONE 6.3 (25%)

BD 1,022.727

Fleet Logistics, Vehicle Bookings, Fuel Reconciliation & TCO Analytics.

📅 Weeks 7–9

MILESTONE 6.4 (25%)

BD 1,022.727

Staff Locker Control, Secure Digital Document Vault & CR Renewal Alerts.

📅 Weeks 10–12

MOD 7 **Module 7: Human Resource Management & Payroll (16 Weeks)**

BD 5,454.548 ▾

MILESTONE 7.1 (25%)

BD 1,363.637

Org Structure, Employee Master, Document Vault & Digital Onboarding.

📅 Weeks 1–4

MILESTONE 7.2 (25%)

BD 1,363.637

Biometric Attendance, Shift Rosters & Bahrain Labour Law Leave System.

📅 Weeks 5–8

MILESTONE 7.3 (25%)

BD 1,363.637

Automated Payroll, SIO/LMRA Compliance, WPS/CBB Export & Loan Advances.

📅 Weeks 9–12

MILESTONE 7.4 (25%)

BD 1,363.637

Performance KPIs, ESS Self-Service Portal, EOSB Gratuity & Production Go-Live.

📅 Weeks 13–16

MOD 8 **Module 8: Hardware Integration & Cloud Cluster (3 Weeks)**

BD 1,022.727 ▾

MILESTONE 8.1 (33.33%)

BD 340.909

3-Tier Server Environment (Dev/Stage/Prod), RabbitMQ & Backup Automation.

📅 Week 1

MILESTONE 8.2 (33.33%)

BD 340.909

Handheld Android QR Scanners, ESC/POS Printers & Biometric Clocks.

📅 Week 2

MILESTONE 8.3 (33.34%)

BD 340.909

Offline Auto-Sync Engine, Multi-Branch Stress Testing & Go-Live.

📅 Week 3

MOD 9 **Module 9: Executive Management Dashboard & BI Cockpit (4 Weeks)**

BD 1,363.636 ▾

MILESTONE 9.1 (25%)

BD 340.909

Data Warehouse OLAP Schema, Non-Blocking ETL Pipelines & KPI Models.

📅 Week 1

MILESTONE 9.2 (25%)

BD 340.909

Commercial, Stock Velocity, Warehouse Verification & Vendor SLA Cockpit.

📅 Week 2

MILESTONE 9.3 (25%)

BD 340.909

Financial Liquidity, VAT, HR Payroll, Fixed Assets & Infra Health Analytics.

📅 Week 3

MILESTONE 9.4 (25%)

BD 340.909

Management Filtering, 7:00 AM WhatsApp/Email Digest & Final Sign-Off.

📅 Week 4

 Dedicated Engineering Team & Transparent Rate Matrix

Full-time dedicated 5-person engineering team rate card in Bahraini Dinars (BHD)

TEAM MONTHLY RATE: BD 1,363.636

All modules are estimated strictly using the agreed dedicated team rate matrix in Bahraini Dinars (BHD). Total dedicated team capacity: 160 hours / month per engineer.

Project Manager & Enterprise Solutions Architect

BD 545.455

BD 136.364 / Wk • BD 3.409 / Hr

100% Dedicated

Back-End Lead Engineer (PHP MVC / REST APIs)

BD 227.273

BD 56.818 / Wk • BD 1.420 / Hr

100% Dedicated

Front-End Lead Engineer (React / UI Design Tokens)

BD 227.273

BD 56.818 / Wk • BD 1.420 / Hr

100% Dedicated

Database & Cloud DevOps Specialist

BD 204.545

BD 51.136 / Wk • BD 1.278 / Hr

100% Dedicated

QA Automation & Multi-Branch Test Lead

BD 159.091

BD 39.773 / Wk • BD 0.994 / Hr

100% Dedicated

 Total Project Investment & Milestone Schedule in Bahraini Dinars (BHD)

Consolidated commercial breakdown and 4-milestone payment tranche distribution for Popular Auto Spare Co.

GRAND TOTAL: BD 34,090.910

Module Reference	Module Domain	Total (BHD)	Tranche 1 (BHD)	Tranche 2 (BHD)	Tranche 3 (BHD)	Tranche 4 (BHD)	Payment Milestone Trigger
MS-001	PCode Generation Module	BD 3,409.091	BD 852.273	BD 852.273	BD 852.273	BD 852.273	Demonstration & Staging Sign-Off
MS-002	Vendor & Purchase Module	BD 4,090.909	BD 1,022.727	BD 1,022.727	BD 1,022.727	BD 1,022.727	Supplier Portal & 3-Way Match Verification
MS-003	Store Verification Module	BD 5,113.636	BD 1,278.409	BD 1,278.409	BD 1,278.409	BD 1,278.409	Scanner Verification & Cycle Count UAT
MS-004	Sales Process & POS Module	BD 5,113.636	BD 1,278.409	BD 1,278.409	BD 1,278.409	BD 1,278.409	Counter POS & Cash Drawer Closing UAT
MS-005	Accounts & VAT Module	BD 4,431.818	BD 1,107.955	BD 1,107.955	BD 1,107.955	BD 1,107.955	GL Double-Entry & 10% VAT Return Sign-Off
MS-006	Enterprise Admin & Fleet	BD 4,090.909	BD 1,022.727	BD 1,022.727	BD 1,022.727	BD 1,022.727	Asset QR Tagging & Fleet TCO Sign-Off
MS-007	HRM & Payroll Module	BD 5,454.548	BD 1,363.637	BD 1,363.637	BD 1,363.637	BD 1,363.637	Biometric Sync & WPS CBB Bank Export UAT
MS-008	Hardware & Cloud Setup	BD 1,022.727	BD 340.909	BD 340.909	BD 340.909	—	3-Tier Cloud & Hardware Peripherals Setup
MS-009	Executive BI Dashboard	BD 1,363.636	BD 340.909	BD 340.909	BD 340.909	BD 340.909	C-Suite Mobile Dashboard & Anomaly Alerts
GRAND TOTAL INVESTMENT (BHD)		BD 34,090.910	BD 8,707.955	BD 8,707.955	BD 8,707.955	BD 7,967.045	35 Verified Milestone Gates



Governance, SLA & Payment Terms

Contractual milestones, UAT acceptance gates, and IP protection protocols

ENTERPRISE SLA

1. Milestone-Driven Invoicing & Payment Tranches

- **Zero Blind Upfront Risk:** Invoices are raised strictly upon completion, demonstration, and staging sign-off of each milestone tranche (25% or 33.3% per gate).
- **15-Day Verification & UAT Grace Period:** The client is entitled to a 15-calendar-day UAT review window per milestone to test all specified acceptance criteria before signing off.
- **Change Request (CR) Protocol:** Any features outside the defined scope are logged via SaNDS Lab Portal and quoted using the transparent hourly rate card (BD 8.523 / Hour).

2. Source Code Ownership, Security & Warranty

- **100% Intellectual Property (IP) Ownership:** Upon settlement of contract invoices, full proprietary source code, DB schemas, and deployment scripts belong exclusively to Popular Auto Spare Co.
- **Zero Third-Party Recurring Royalties:** Built on open-standard enterprise stacks (PHP 8.3 MVC, MySQL InnoDB, React 18, RabbitMQ, SQLite) with zero vendor lock-in.
- **12-Month Post-Launch Bug Warranty:** 365 days of comprehensive bug-fixing SLA and security patch support following the final production deployment of each module.

Stakeholder Authorization & Digital Sign-Off Console

By affixing digital signatures and authorizations below, all participating executive and technical stakeholders ratify the deliverables, implementation timeline, SLA commitments, and financial disbursement gates for **Master 9-Module ERP Milestone & Budget Roadmap (SL-POP-ERP-SUMMARY-001)**:

1. Engineering Preparation & Stakeholder Verification Matrix

DEVELOPER / ARCHITECT Ancy Varghese Thekkan Software Architect, SaNDS Lab PREPARED (25-SEPTEMBER-2026) COMPLETED	VERIFIED BY (DEVELOPER) Ajit Kumar KV CEO & Managing Director, SaNDS Lab VERIFIED (30-SEPTEMBER-2026) CHECKED & VERIFIED	CONSULTANT REVIEW UniGlobal Business Solutions Project Management Consultant In Active Review IN REVIEW	CLIENT APPROVAL Popular Auto Spare & A/C Parts Authorized Executive Signatory Pending Approval PENDING SIGNATURE
---	---	--	---

2. Executive Multi-Party Digital Acceptance Console

Popular Auto Spare Co. W.L.L Client Executive Sign-off & Commercial Approval No Signature on Record PENDING SIGN-OFF	SaNDS Lab Middle East W.L.L Lead Solutions Architect & Engineering Director No Signature on Record PENDING SIGN-OFF	UniGlobal Consultancy Executive Strategic Advisor & Governance Lead No Signature on Record PENDING SIGN-OFF
--	---	---